Settle complexity of min-cost popular maximum matchings in many-to-one settings

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Website #36 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Given a hospitals/residents instance $G = (R \cup H, E)$ with strict (possibly incomplete) preferences, hospital capacities $c_h := \mathrm{cap}(h) \geq 1$, and edge-costs $\mathrm{cost} : E \to \mathbb{R}$, define a maximum matching to be a feasible many-to-one matching of maximum cardinality. For two matchings M, N, residents vote in the usual way, and each hospital h compares $M(h)$ to $N(h)$ by removing common residents, padding the smaller side with dummies ranked below all real residents, and then taking the *minimum* over bijections of the sum of pairwise votes. The net margin is

$$\Delta(M, N) = \sum_{v \in R \cup H} \mathrm{vote}_v(M, N).$$

A *popular maximum matching* is a maximum matching M such that $\Delta(M, N) \geq 0$ for every maximum matching N.

The optimization problem asks for a minimum-cost matching in this family, or to decide that no such matching exists. For complexity statements, I interpret costs as rational numbers encoded in binary.

Alternative readings considered

The only ambiguity is the use of real costs in a complexity question. I adopt the standard Turing-model reading: edge costs are rationally encoded. I also treat the feasible family as possibly empty; thus the associated decision version is:

$$\exists \text{ a popular maximum matching } M \text{ with } \mathrm{cost}(M) \leq K\ ?$$

Related literature found

- **Brandl–Kavitha (2017)** [`BACKGROUND/TOOL`]: introduces the *strong* capacitated popularity notion used here, where a capacitated vertex compares symmetric differences by the most adversarial bijection. They give a linear-time algorithm for a maximum-size popular matching among *all* matchings in the many-to-many model, but not for popularity restricted to maximum matchings and not for min-cost optimization. (drops.dagstuhl.de)
- **Kavitha (2021)** [`TOOL/PARTIAL`]: solves the one-to-one analogue: in a marriage instance, min-cost popular maximum matching is polynomial-time solvable, and popular maximum matchings are characterized by the absence of positive alternating cycles and positive alternating paths from unmatched vertices. This is the closest one-to-one template for the present problem. (drops.dagstuhl.de)

- **Kavitha–Makino (2024/25)** [`PARTIAL/TOOL`]: solves two nearby many-to-one problems under the same strong hospital-voting rule: (i) min-cost popular matching for complete preferences, and (ii) min-cost popular *perfect* matching for arbitrary incomplete preferences. Their paper also explicitly points out the min-cost popular maximum matching problem in the many-to-one setting with incomplete preferences as an open problem. (tcs.tifr.res.in)

- **Nasre–Rawat (2016)** [`PARTIAL`]: computes a matching popular among maximum-cardinality matchings in a generalized HR setting, but under a *weaker* hospital comparison rule (identity correspondence rather than adversarial minimum). Hence it does not settle the present model. The mismatch of voting notions is explicitly noted by Kavitha–Makino. (arxiv.org)

- **Nasre–Nimbhorkar–Ranjan–Sarkar (2021)** [`PARTIAL/BACKGROUND`]: studies popularity among feasible matchings in HR with two-sided lower quotas/criticality. This is structurally close, but the feasible family is different from "all maximum-cardinality matchings" and the paper does not address min-cost optimization over our family. (drops.dagstuhl.de)

Proof sketch

I do *not* settle the full optimization problem. Instead I prove a structural characterization and a polynomial-time *recognition* theorem for a *given* candidate matching M, then derive two unconditional polynomial-time subclasses from the literature.

The key observation is that the proof of Theorem 7 in Kavitha–Makino for *popular* matchings in complete-preference HR instances uses completeness only once: to justify that it suffices to compare M against *maximum* challengers. Once the feasible family is *already* the set of maximum matchings, that completeness hypothesis is no longer needed. The rest of the weighted alternating-cycle/path argument goes through unchanged.

So I move to the standard clone graph G', fix any realization M' of a maximum matching M, and consider the restricted subgraph $G'_{M'}$ in which unmatched original edges keep *all* hospital copies but each matched original edge keeps only its realized copy. On this graph I put the usual edge weights $\mathrm{wt}_{M'} \in \{-2, 0, 2\}$, where $\mathrm{wt}_{M'}(e)$ is the sum of the endpoint votes for using e instead of their assignments in M'.

Then:

- if there is a positive M'-alternating cycle, toggling it yields another maximum matching that defeats M;

- if there is a positive M'-alternating path with an unmatched endpoint and with endpoints not two clones of the same hospital, toggling it again yields another maximum matching that defeats M;

- conversely, from any maximum challenger N, one can realize N inside $G'_{M'}$ using the hospital bijections that attain the adversarial hospital votes in the M-vs-N election, and then the symmetric difference decomposes into alternating cycles and even alternating paths. The contribution of each component is exactly controlled by its $\mathrm{wt}_{M'}$-weight.

This gives a clean forbidden-substructure characterization for popular maximum matchings in arbitrary incomplete HR instances.

A second step contracts the matched edges of M' to form a directed weighted graph. In that digraph, positive alternating cycles become positive directed cycles, and positive alternating paths become positive directed source/target paths. Thus *recognizing* whether a given maximum matching is popular maximum is polynomial-time.

This is weaker than solving the optimization problem, but it is a real complexity statement: the threshold decision problem lies in NP, and the quantifier "for all maximum matchings N" admits a succinct polynomially checkable certificate. Finally, by invoking known results, I recover polynomial-time solvability when all capacities are 1 (the marriage case) and when the HR instance admits a perfect matching.

Main result

Theorem 1. *Let $G = (R \cup H, E)$ be a hospitals/residents instance with strict preferences (possibly incomplete), and let M be a maximum matching in G.*

For each hospital h, create clones $h^1, \dots, h^{c_h}$, where $c_h = \mathrm{cap}(h)$. Let $G' = (R \cup H', E')$ be the corresponding clone graph, where each clone h^i inherits h's preference order, and each resident orders the clones of a hospital as

$$h^1 \succ h^2 \succ \cdots \succ h^{c_h}.$$

Fix any *realization M' of M in G'. Define the subgraph*

$$E'_{M'} := M' \,\cup\, \{(r, h^i) : (r, h) \in E \setminus M,\ 1 \le i \le c_h\}, \qquad G'_{M'} := (R \cup H', E'_{M'}).$$

For $e = (r, h^i) \in E'_{M'}$, define

$$\mathrm{wt}_{M'}(e) = \mathrm{vote}_r(h^i, M'(r)) + \mathrm{vote}_{h^i}(r, M'(h^i)) \in \{-2, 0, 2\},$$

where unmatched is treated as the worst option.

Then M is a popular maximum matching in G if and only if the following hold in $G'_{M'}$:

1. *there is no M'-alternating cycle C with $\mathrm{wt}_{M'}(C) > 0$;*
2. *there is no M'-alternating path ρ with an unmatched endpoint, whose two endpoints are not clones of the same hospital, and with $\mathrm{wt}_{M'}(\rho) > 0$.*

Consequently, recognition of a given popular maximum matching is polynomial-time. Hence the cost-threshold decision version of the optimization problem belongs to NP.

Proof

Lemma 1 (Sufficiency)**.** *Let M be a maximum matching and let M' be a realization satisfying conditions (1)–(2) of the theorem. Then M is a popular maximum matching.*

Proof. Fix any maximum matching N in G. I will construct a realization N' of N inside $G'_{M'}$.

For every edge $(r, h) \in M \cap N$, if M' contains (r, h^i), put the same edge (r, h^i) into N'.

Now fix a hospital h and consider $(N(h) \setminus M(h))$. In the evaluation of $\mathrm{vote}_h(M, N)$, after removing common residents and padding the smaller side with dummies, each resident $r \in N(h) \setminus M(h)$ is compared by h either with some resident $r' \in M(h) \setminus N(h)$ or with a dummy $\perp$. For each such r:

- if r is compared with a real resident r', and M' contains (r', h^i), put (r, h^i) into N';

- if r is compared with a dummy, choose any clone h^i that is unmatched in M' and not yet used for N', and put (r, h^i) into N'.

Doing this independently for each hospital produces a matching $N' \subseteq E'_{M'}$. By construction N' is a realization of N, so $|N'| = |N| = |M| = |M'|$.

Hence every component of $M' \oplus N'$ is either an alternating cycle or an even alternating path.

Claim 1. If ρ is an alternating path in $M' \oplus N'$, then its two endpoints are not clones of the same hospital.

Proof of Claim 1. Fix a hospital h.

If $|M(h)| \leq |N(h)|$, then every clone of h that is matched in M' is also matched in N': residents common to M and N keep their clones, and each resident in $N(h) \setminus M(h)$ takes either the clone of the resident it is paired against or an unmatched clone. So a clone of h that is unmatched in N' must already have been unmatched in M'.

If $|M(h)| > |N(h)|$, then every clone of h that is unmatched in M' remains unmatched in N', because N' uses strictly fewer clones of h than M'.

Thus an alternating path cannot have one endpoint a clone of h unmatched in M' and the other endpoint another clone of h unmatched in N'. This proves the claim. $\square$

I now compute the contribution of each component of $M' \oplus N'$ to $\Delta(M, N)$.

For any unmatched edge $e = (r, h^i) \in E'_{M'} \setminus M'$, the quantity $\mathrm{wt}_{M'}(e)$ is exactly the sum of the two endpoint votes for N versus M when r is assigned to h (equivalently, to h^i) instead of its partner in M'. Matched edges of M' have weight 0.

Claim 2. If C is an alternating cycle in $M' \oplus N'$, then the total contribution of the vertices of C to $\Delta(M, N)$ equals $-\mathrm{wt}_{M'}(C)$.

Proof of Claim 2. Along C, every vertex is incident to exactly one edge of $N' \setminus M'$, and each such edge contributes to $\mathrm{wt}_{M'}(C)$ exactly the votes of its two endpoints for N versus M. Therefore

$$\sum_{v \in V(C)} \mathrm{vote}_v(N, M) = \mathrm{wt}_{M'}(C).$$

Since $\mathrm{vote}_v(M, N) = -\mathrm{vote}_v(N, M)$, the contribution to $\Delta(M, N)$ is $-\mathrm{wt}_{M'}(C)$. $\square$

Claim 3. If ρ is an alternating path in $M' \oplus N'$, then the total contribution of the vertices of ρ to $\Delta(M, N)$ equals

$$-\big(\mathrm{wt}_{M'}(\rho) - 1\big).$$

Proof of Claim 3. Because $|M'| = |N'|$, an alternating path in $M' \oplus N'$ has one endpoint unmatched in M' and one endpoint unmatched in N'.

All vertices of ρ except the endpoint unmatched in N' are incident to exactly one edge of $N' \setminus M'$, hence are counted in $\mathrm{wt}_{M'}(\rho)$. The endpoint unmatched in N' contributes $+1$ to $\Delta(M, N)$ (it prefers being matched in M to being unmatched in N) but contributes nothing to $\mathrm{wt}_{M'}(\rho)$. Therefore

$$\sum_{v \in V(\rho)} \mathrm{vote}_v(M, N) = -\mathrm{wt}_{M'}(\rho) + 1 = -\big(\mathrm{wt}_{M'}(\rho) - 1\big).$$

$\square$

Summing Claim 2 over all alternating cycles and Claim 3 over all alternating paths in $M' \oplus N'$, we obtain

$$\Delta(M, N) = -\sum_{C} \mathrm{wt}_{M'}(C) - \sum_{\rho} \big(\mathrm{wt}_{M'}(\rho) - 1\big).$$

By hypothesis, every alternating cycle has weight ≤ 0. By Claim 1, every alternating path appearing in $M' \oplus N'$ has endpoints not clones of the same hospital, so hypothesis (2) gives $\mathrm{wt}_{M'}(\rho) \leq 0$ for every such path. Hence every summand on the right-hand side is nonnegative, so $\Delta(M, N) \geq 0$.

Since N was an arbitrary maximum matching, M is popular among maximum matchings. $\square$

Lemma 2 (Necessity, for every realization)**.** *Let M be a popular maximum matching. Then* every *realization M' of M satisfies conditions (1)–(2) of the theorem.*

Proof. Fix any realization M' of M.

Suppose first that condition (1) fails: there is an alternating cycle C in $G'_{M'}$ with $\mathrm{wt}_{M'}(C) > 0$. Let

$$N' := M' \oplus C.$$

Then $|N'| = |M'|$. Identifying clones of the same hospital yields a matching N in G with $|N| = |M|$, hence N is also maximum.

For each hospital h appearing on C, the cycle itself defines a bijection between $M(h) \setminus N(h)$ and $N(h) \setminus M(h)$: the resident removed from a clone of h in M' is paired with the resident inserted into that same clone in N'. Under this particular bijection, the total vote of the vertices on C for N versus M is exactly $\mathrm{wt}_{M'}(C)$. Since $\mathrm{vote}_h(M, N)$ is the *minimum* over all bijections, the true contribution of hospital h to $\Delta(M, N)$ is at most the contribution under this cycle-induced bijection. Therefore

$$\sum_{v \in V(C)} \mathrm{vote}_v(M, N) \leq -\,\mathrm{wt}_{M'}(C) < 0.$$

All vertices outside C have the same assignments in M and N, so they contribute 0. Hence $\Delta(M, N) < 0$, contradicting popularity of M among maximum matchings.

Now suppose condition (2) fails: there is an alternating path ρ in $G'_{M'}$ with an unmatched endpoint, with endpoints not clones of the same hospital, and with $\mathrm{wt}_{M'}(\rho) > 0$. Let

$$N' := M' \oplus \rho.$$

Because ρ is an even alternating path, $|N'| = |M'|$; projecting N' back to G gives a maximum matching N.

As above, ρ defines a concrete comparison between the hospital-side symmetric differences for every hospital touched by ρ. The endpoint condition ensures this comparison is legitimate in the original HR instance: if the endpoints were two clones of the same hospital, the clone-level comparison would contain a real-vs-dummy and a dummy-vs-real comparison that cancel when one passes back to the true hospital vote in G; this is exactly the pathology excluded here. Compare the discussion in Kavitha–Makino around this point. (tcs.tifr.res.in)

Under the path-induced comparison, the total vote for N versus M on the vertices of ρ, except for the endpoint unmatched in N', is $\mathrm{wt}_{M'}(\rho)$. The endpoint unmatched in N' contributes one vote to M. Hence

$$\sum_{v \in V(\rho)} \mathrm{vote}_v(M, N) \leq -\big(\mathrm{wt}_{M'}(\rho) - 1\big).$$

Because every edge-weight is in $\{-2, 0, 2\}$, the path-weight is even; thus $\mathrm{wt}_{M'}(\rho) > 0$ implies $\mathrm{wt}_{M'}(\rho) \geq 2$, so the right-hand side is at most -1. Therefore $\Delta(M, N) < 0$, again contradicting popularity of M.

Thus both conditions must hold for every realization M'. $\square$

Proof of the theorem. The forward direction is Lemma 2, the reverse direction is Lemma 1. $\square$

Corollary 1 (Polynomial-time recognition). *Given G and a matching M, one can test in polynomial time whether M is a popular maximum matching.*

Proof. First verify that M is maximum, using any polynomial-time algorithm for bipartite b-matching.

Now fix an arbitrary realization M'. Since capacities larger than the degree are irrelevant, we may first replace each c_h by $\min\{c_h, \deg(h)\}$; thus the total number of clones is at most $|E|$, so $G'_{M'}$ has polynomial size.

Contract every matched edge $(r, h^i) \in M'$ to a directed-state vertex x_{r,h^i}. For every unmatched edge $e = (r, h^j) \in E'_{M'} \setminus M'$:

- if r is matched in M' to h^i and h^j is matched in M' to r', add the directed arc

$$x_{r,h^i} \to x_{r',h^j}$$

  with weight $\mathrm{wt}_{M'}(e)$;

- if r is unmatched in M' and h^j is matched to r', add a source s_r and an arc

$$s_r \to x_{r',h^j}$$

  of weight $\mathrm{wt}_{M'}(e)$;

- if r is matched to h^i and h^j is unmatched in M', add a sink t_{h^j} and an arc

$$x_{r,h^i} \to t_{h^j}$$

  of weight $\mathrm{wt}_{M'}(e)$.

An alternating cycle in $G'_{M'}$ corresponds exactly to a directed cycle in this digraph, with the same total weight, because matched edges contribute weight 0. Likewise, an alternating path with an unmatched resident endpoint corresponds to a directed path from some s_r to some state vertex, and an alternating path with an unmatched clone endpoint corresponds to a directed path from some state vertex to some t_{h^j}. The forbidden "same hospital" case is checked locally at the terminal pair (x_{r,h^i}, t_{h^j}).

Hence:

- condition (1) is violated iff the digraph has a positive directed cycle;
- condition (2) is violated iff, after ruling out positive cycles, there is a positive directed path of one of the two allowed source/target types.

Positive directed cycles are detected in polynomial time by negating arc weights and running any negative-cycle detection algorithm. If no positive cycle exists, then after negation there is no negative cycle, so maximum path weights from the allowed sources (or to the allowed sinks, by reversing the digraph) are obtained from shortest-path distances. Therefore both conditions are polynomial-time checkable.

By the theorem, M is a popular maximum matching iff these two checks pass. □

Corollary 2 (Decision version is in NP). *Assume the edge costs are rational numbers encoded in binary. Then*

$$\{(G, K) : \exists \textit{ a popular maximum matching } M \textit{ with } \mathrm{cost}(M) \le K\} \in \mathrm{NP}.$$

Proof. Guess the matching M. Its feasibility, cardinality, and cost are checked in polynomial time, and by the previous corollary its popularity among maximum matchings is checked in polynomial time. □

Corollary 3 (Two polynomial-time subclasses)**.** *The optimization problem is polynomial-time solvable in each of the following cases:*

1. *every hospital has capacity* 1*;*
2. *G admits a perfect matching (i.e. every resident can be matched and every hospital can be filled simultaneously).*

Proof. If every hospital has capacity 1, we are in the marriage setting. Kavitha proved that for any marriage instance with strict preferences and edge costs, a min-cost popular maximum matching can be computed in polynomial time. Hypothesis audit: our case (1) is exactly a marriage instance; preferences are strict and costs are given on edges. Hence the cited theorem applies directly. (drops.dagstuhl.de)

If G admits a perfect matching, then every maximum matching has size $|R| = \sum_{h \in H} c_h$, so every maximum matching is perfect. Therefore "popular maximum" is exactly "popular perfect". Kavitha–Makino proved that in any HR instance with strict preferences (possibly incomplete) and edge costs, if the instance admits a perfect matching, then a min-cost popular perfect matching can be computed in polynomial time. Hypothesis audit: case (2) matches these assumptions exactly. (tcs.tifr.res.in) □

Gap to the full statement

I have *not* proved a polynomial-time algorithm for the general optimization problem, and I have *not* proved NP-hardness either.

What *is* settled above:

- a structural characterization of popular maximum matchings in arbitrary incomplete HR instances;
- polynomial-time recognition of a *given* candidate matching;
- membership of the threshold decision problem in NP;
- polynomial-time solvability in the one-to-one case and in the perfect-matchable many-to-one case.

What remains open in my work is the global optimization step: minimizing cost over the P-recognizable family of popular maximum matchings in a general many-to-one instance. This is also the exact many-to-one incomplete-preference problem explicitly singled out as open by Kavitha–Makino. (tcs.tifr.res.in)

Self red-team

- **Counterexample attempt 1: the Section 1.1 clone-pathology example.** Kavitha–Makino exhibit a many-to-one matching that is popular in the HR instance although one realization is not popular in the full clone graph. This does *not* break my theorem, because my characterization is in the restricted graph $G'_{M'}$, not in the full clone graph G', and it asks only for absence of positive alternating cycles/paths, not for popularity of the realization as a one-to-one matching. (tcs.tifr.res.in)

- **Counterexample attempt 2: omit the endpoint condition in (2).** This fails. If an alternating path has endpoints that are two clones of the same hospital, the clone-level comparison creates a spurious real-vs-dummy / dummy-vs-real pair that disappears in the true hospital comparison in G. So the exclusion is genuinely necessary. (tcs.tifr.res.in)
- **Counterexample attempt 3: boundary cases.**
  - If M is perfect, there are no unmatched vertices, so condition (2) disappears; this matches the known popular-perfect characterization.
  - If all capacities are 1, the clone graph is the original graph and the theorem reduces to the one-to-one forbidden cycle/path criterion.

  Both checks are consistent. (tcs.tifr.res.in)

The three likeliest referee objections and my responses are:

1. *"You secretly used complete preferences."* No. The only place where completeness entered the 2024 argument was the sentence "it suffices to compare against maximum matchings." Here that restriction is built into the definition of popular maximum matching, so the rest of the weighted alternating-structure proof goes through unchanged.
2. *"Why may you choose an arbitrary realization M'?"* Because necessity was proved for *every* realization, while sufficiency needs only *one*. Hence for a fixed matching M, either all realizations satisfy the criterion or none do.
3. *"Is the recognition algorithm really polynomial?"* Yes. After truncating irrelevant oversized capacities, the clone graph is polynomial in the input size, and the forbidden structures reduce to positive cycles and source/target paths in a directed weighted graph. Those are polynomial-time detectable by negative-cycle detection and shortest paths after weight negation.

Ideas tried

1. **Completion-to-perfect-matching gadget.** Idea: add dummy residents/hospitals so that every maximum matching extends canonically to a perfect matching, then apply the many-to-one popular-perfect machinery. Obstruction: it is surprisingly hard to make the added dummy *voters* neutral under the adversarial hospital-bijection rule. Naive gadgets add extra hospital and dummy-resident votes that do not cancel cleanly.
2. **Direct reduction to a one-to-one clone instance.** Idea: clone each hospital into c_h copies and solve a one-to-one popular maximum matching problem there. Obstruction: already ruled out by the many-to-one example in Kavitha–Makino, where the HR instance has popular maximum matchings that are not images of popular maximum matchings in the naive clone instance. The surjection that exists for popular *perfect* matchings fails here. (tcs.tifr.res.in)
3. **Reduction to lower quotas / critical matchings.** Idea: encode "being maximum" as satisfying lower quotas or matching a critical set, then use existing critical/feasible-popularity

algorithms. Obstruction: maximum *edge-cardinality* in HR is not naturally the same as matching a fixed vertex set in the original many-to-one instance. Slot-level cloning reintroduces the same realization/bijection difficulty.

4. **Polyhedral lift from recognition.** Idea: since I obtained a polynomial-time recognition theorem, maybe optimize by separation over the convex hull. Obstruction: recognizing *integral* feasible points is much weaker than separating the convex hull of all popular maximum matchings. I did not find a compact linear description or an extension of the one-to-one dual-certificate machinery that survives the many-to-one adversarial hospital vote.

Invoked results

1. **Kavitha (2021), Theorem 3.** *Statement.* Given a marriage instance $G = (A \cup B, E)$ with strict preferences and an edge-cost function $c : E \to \mathbb{R}$, a min-cost popular maximum matching can be computed in polynomial time. *Classification.* `TOOL`. *Citation.* Telikepalli Kavitha, *Maximum Matchings and Popularity*, ICALP 2021, DOI 10.4230/LIPIcs.ICALP.2021.85. (drops.dagstuhl.de) *Hypothesis audit.* In Corollary 3(1), every hospital has capacity 1, so our instance is exactly a marriage instance; preferences are strict by hypothesis; costs are part of the input. Hence the theorem applies verbatim.

2. **Kavitha–Makino (2024/25), Theorem 6.** *Statement.* Let $G = (R \cup H, E)$ be a hospitals/residents instance with strict preferences (possibly incomplete) and edge costs. If G admits a perfect matching, then a min-cost popular perfect matching can be computed in polynomial time. *Classification.* `TOOL`. *Citation.* Telikepalli Kavitha and Kazuhisa Makino, *Min-Cost Popular Matchings in a Hospitals/Residents Instance with Complete Preferences*; the perfect-matching subroutine is stated for incomplete preferences as Theorem 6. (tcs.tifr.res.in) *Hypothesis audit.* In Corollary 3(2), our instance has strict preferences and edge costs, and we assume it admits a perfect matching. In that situation every maximum matching is perfect, so optimizing over popular maximum matchings is the same as optimizing over popular perfect matchings. Thus the cited theorem applies exactly.

Self-confidence

3 — I am confident in the structural theorem, the recognition algorithm, and the two tractable subclasses, but I did not settle the full optimization complexity.